Larger budgets, growing ranks, and pointwise tensor inference

September 12, 2026

Status: 2026-09-12. This supplement gives complete supplied derivations in specific regimes of the remaining problems. It does not assert a general polynomial weak-signal minimax algorithm. Previous stages are preserved.

Observe $Z = A + \tau G$ with independent standard Gaussian tensor entries, known $\tau > 0$, norm bound $\|A\|_F \le M$, and supplied Tucker or TT caps. Unless specified otherwise, dimensions are equal to p and $K \ge 3$. For $L = p^{-1/2}\sum_{i=1}^{p} E_i$, write $\psi(A) = \langle L, A\rangle$, where $E_i = e_i^{\otimes K}$. The criteria $\mathcal{L}_0$ and $\mathcal{L}_{\max}$ are optimal expected length at zero and optimal worst-case expected length among intervals honest over the whole stated class. Fix $0 < \alpha < 1/4$.

G1. Sharp growing-cap rates with a polynomial algorithm

Let $r = \min(p, \min_k \bar{r}_k)$, where the minimum is over the supplied constrained unfoldings (Tucker modes or TT cuts). Suppose $r \ge c\sqrt{p}$ for a fixed $c > 0$. Uniformly over all remaining caps, all p, and every $M, \tau > 0$,

$$\mathcal{L}_0 \asymp_{\alpha,c} \mathcal{L}_{\max} \asymp_{\alpha,c} \min\{M\sqrt{r/p}, \tau\}.$$

Minimax squared risk has the square of this rate. The constants do not depend on growing ranks or tensor order. Choose deterministically between the structural interval $[-h_0, h_0]$, $h_0 = M\sqrt{r/p}$, and the direct Gaussian interval centered at $\psi(Z)$ with radius $z_{1-\alpha/2}\tau$. This takes $O(p)$ accessed entries and arithmetic for the coordinate loading. For a rotated flat loading, obtaining the contractions has its additional polynomial input-access cost. No factor estimation or exponential scan is required.

Upper proof. An unfolding attaining the smallest cap has loading matrix singular values $1/\sqrt{p}$, repeated p times. Matrix singular-value duality and Cauchy–Schwarz bound its contraction with any rank-at-most-r matrix of norm M by $M\sqrt{r/p}$. Thus the structural interval covers surely. The direct error is $N(0, \tau^2)$. Deterministic radius selection proves the length upper bound. Choosing estimator zero when $h_0 \le \tau$ and the direct estimator otherwise gives squared risk at most $\min(h_0^2, \tau^2)$.

Lower proof. For a uniform size-r subset S of $\{1, \dots, p\}$ set

$$A_S = b\sum_{i\in S} E_i, \qquad b = \min\{M/\sqrt{r}, \tau\sqrt{\log(1 + p/(4r^2))}\}.$$

All constrained unfolding ranks are r, the norm is at most M, and the common target is $\Delta = rb/\sqrt{p}$. For independent supports S, T, the full Gaussian likelihood second moment equals $\mathbb{E}\exp\{b^2|S\cap T|/\tau^2\}$. To bound it, for $H = |S \cap T|$ factorial moments give $\mathbb{E}\binom{H}{j} = \binom{r}{j}(r)_j/(p)_j \le \binom{r}{j}(r/p)^j$. Expand $(1+t)^H$ at $t = e^{b^2/\tau^2} - 1$ to obtain

$$1 + \chi^2(Q, P_0) \le (1 + rt/p)^r \le \exp(r^2 t/p) \le e^{1/4}.$$

Put $x = p/(4r^2) \le 1/(4c^2)$. Since $x/(1+x) \le \log(1+x) \le x$,

$$\frac{\tau}{2\sqrt{1 + 1/(4c^2)}} \le \frac{\tau r}{\sqrt{p}}\sqrt{\log(1 + p/(4r^2))} \le \frac{\tau}{2}.$$

Consequently $\Delta \asymp_c \min(M\sqrt{r/p}, \tau)$. Total variation is at most $\eta = \sqrt{e^{1/4}-1}/2 < 1/2$. Coverage of both zero and the common target implies $\mathbb{E}_0|C| \geq \Delta(1-2\alpha-\eta)$, which is positive for $\alpha < 1/4$. The nearest-target testing argument also gives squared risk at least $\Delta^2(1-\eta)/8$. This proves all claims.

This closes the large-cap regime using the earlier sparse-diagonal lower bound. It is a consequence of that bound with a newly matched upper bound, not a new sparse-mixture technique. Under $M = \tau\sqrt{p}$ the rate is τ, so the vanishing fixed-cap improvement has already disappeared by rank order $\sqrt{p}$, well below unrestricted rank p.

G2. General adaptation across positive growing caps is impossible

Set $M = \tau\sqrt{p}$ and let the outer smallest cap satisfy $r \geq c\sqrt{p}$. Every interval honest on that outer class has $\mathbb{E}_0|C| \geq c_{\alpha,c}\tau$. For a fixed positive inner cap vector, the previously established oracle minimax rate is

$$\mathfrak{r}_K(p) = \frac{\tau}{\sqrt{p}} \begin{cases} (\log(ep))^{3/2}, & K = 3, \\ \sqrt{\log(ep)}, & K \geq 4. \end{cases}$$

Zero belongs to every such inner class. Its worst-case expected length under outer honesty is therefore at least order τ, a diverging factor $\sqrt{p}/(\log(ep))^{3/2}$ or $\sqrt{p/\log(ep)}$ larger than its known-class oracle rate. No algorithm, including an exponential one, can be simultaneously minimax in rate over these positive fixed and growing cap classes. This is stronger than excluding only the known-rank-zero oracle. It does not conflict with earlier adaptation across *fixed* outer caps, whose constants were allowed to depend on that outer cap.

Nonzero version. Suppose A_0 is supported on the first q coordinates in every mode, with norm $a < M$ and actual constrained ranks $r_{0,k}$. Let $m = p - q$, $s = \min(m, \min_k(\bar{r}_k - r_{0,k}))$ and $M_0 = \sqrt{M^2 - a^2}$. If $s \geq c\sqrt{m}$, then

$$\mathbb{E}_{A_0}|C| \gtrsim_{\alpha,c} \min\{M_0\sqrt{s/p}, \tau\sqrt{m/p}\}.$$

Proof. Add the G1 alternatives on the disjoint remaining block. Norm squares add; every unfolding rank increases by at most s. Translation by A_0 preserves Gaussian likelihood overlaps, and normalization of the target multiplies the dimension-m lower bound by $\sqrt{m/p}$. This gives a positive nonzero-base obstruction when weak extra rank fits inside the supplied caps. Arbitrarily oriented versions follow by using orthogonal product blocks when the transformed loading is flat on that block.

W1. A dyadic loading bound using the full weight profile

Return to a prescribed orthogonal product loading $L_w = \sum_{i=1}^m w_i U_i$, with decreasing absolute weights and fixed positive caps. The ambient mode dimensions may exceed m. Partition the ordered indices into $B_0 = \{1\}$ and $B_j = \{2^{j-1}+1, \ldots, \min(2^j, m)\}$ for $j \geq 1$, stopping when empty. Let $a_j = \max_{i \in B_j} |w_i|$, $d_j = |B_j|$, and $\alpha_j = \alpha/2^{j+1}$. Define

$$H_K(w) = \sum_j a_j (j+1)^{\nu_K}, \qquad \nu_3 = 3/2, \quad \nu_K = 1/2 \ (K \geq 4).$$

There is a full-data honest interval satisfying the deterministic bound

$$|C_w| \leq C_{K,\bar{r},\alpha} \min\{M\|w\|_\infty, \tau H_K(w)\}.$$

The minimax squared risk is at most a constant times the square of the same expression. This is a statistical upper bound using the ideal block residual fits, not a polynomial implementation of those fits.

Proof. Rotate the known orthogonal factors into coordinates. Restrict the data to each d_j^K block; this cannot increase its norm or rank caps. Apply the earlier weighted residual confidence projection at error α_j. With $J_2 = J(2\bar{r})$, its block diameter is at most

$$4J_2^{K/2} a_j \sum_{s=1}^{d_j} \lambda_s^{(j)} s^{-K/2},$$

where

$$\lambda_s^{(j)} = \tau \sqrt{2\{s \log(ed_j/s) + Ks \log(1+4K) + \log(2d_j/\alpha_j)\}}.$$

For $s \geq 1$, $\log(ed_j)$ and $\log(1/\alpha_j)$ are bounded by a fixed constant times $j+1$. Thus $\lambda_s^{(j)} \leq C_{K,\alpha} \tau \sqrt{s(j+1)}$. The remaining sum is harmonic for $K = 3$, bounded by $C(j+1)$, and bounded by a K-dependent constant for $K \geq 4$. Add the block intervals and use $\sum_j \alpha_j \leq \alpha$. Their lengths sum to at most $C\tau H_K(w)$. The old structural radius is at most $M\sqrt{\min \bar{r}_k}\|w\|_\infty$, which is $O(M\|w\|_\infty)$ for fixed caps. Select the smaller deterministic bound before seeing the data.

For squared error use block residual fits with error budgets $\beta_j = 2^{-j-4}$ and deterministic optimization tolerance τ. The earlier excess-tail proof gives each block error an L_2 norm at most $C\tau a_j (j+1)^{\nu_K}$: the deterministic diameter term has the bound just derived, and the excess term has L_2 norm $O(\tau a_j)$ because $\sum_s s^{-K/2} < \infty$ and $\mathbb{E}T_{0,j}^2 \leq 2\beta_j \tau^2$. Minkowski's inequality bounds the L_2 norm of their summed error by $C\tau H_K(w)$, without requiring independence. Compare with estimator zero and its structural risk bound. The finite number of blocks makes measurability and summation immediate.

W2. Sharp rates at every budget for polynomially decreasing loadings

Suppose for fixed $\beta > 0$ and $C_0 < \infty$ that the nonzero, decreasing weights satisfy

$$|w_i| \leq C_0 |w_1| i^{-\beta}, \qquad i = 1, \ldots, m.$$

For fixed positive caps and order, uniformly over all $m, M, \tau > 0$,

$$\mathcal{L}_0(\Theta; L_w) \asymp \mathcal{L}_{\max}(\Theta; L_w) \asymp |w_1| \min(M, \tau),$$

and minimax squared risk is $\asymp |w_1|^2 \min(M, \tau)^2$. Constants may depend on $\beta, C_0, K, \bar{r}, \alpha$, but not on dimension or budget. This includes all signed power-law weights, whether normalized or not. In particular, $0 < \beta < 1/2$ permits diffuse unit-Frobenius loadings with $|w_1| \to 0$; the result is not limited to a constant product overlap with a unit-Frobenius loading.

Proof. For $j \geq 1$, $a_j \leq C_0 |w_1| 2^{-\beta(j-1)}$. The series $\sum_j 2^{-\beta(j-1)}(j+1)^{\nu_K}$ is finite for every fixed $\beta > 0$. W1 therefore gives the matching length and risk upper bounds. For the lower bounds compare zero with tU_1, where $t = \min(M, \tau)/2$ and its sign agrees with w_1. These rank-one means belong to every positive cap class, their target difference is $t|w_1|$, and their total variation is at most $2\Phi(1/4) - 1 < 1/4$. The common-target interval argument and two-point squared-loss argument give the asserted lower orders. No lower power-envelope assumption on the other weights is needed.

As $\beta \downarrow 0$ the constants deteriorate: the dyadic sum is of order at most $1 + \beta^{-(\nu_K+1)}$ up to fixed constants. Thus this theorem does not cover a dimension-dependent exponent approaching zero uniformly, and does not contradict the extra logarithmic factors for flat loadings. Completely arbitrary unequal weight sequences, including many slowly varying profiles, remain uncharacterized.

Nonorthogonal perturbations. Suppose $L = L_w + R$, with the same weight condition and a known otherwise arbitrary tensor R satisfying $\|R\|_F \leq \epsilon |w_1|$ for a fixed $0 \leq \epsilon < 1$. The same sharp rates hold with constants depending also on $1 - \epsilon$. For the lower bound, $|\langle L, U_1 \rangle| \geq (1 - \epsilon)|w_1|$. For large budgets add a direct interval for $\langle R, A \rangle$ at a separate error allocation, costing $O(\tau \|R\|_F)$. For small budgets, $\rho(L_w) = |w_1|$ by generalized Hölder and $\rho(R) \leq \|R\|_F$; the canonical expansion bounds the structural radius by $\sqrt{J} M(1+\epsilon)|w_1|$. The same reasoning applies to squared error, using the direct residual estimator and Minkowski. This gives an all-budget result for a class of loadings without an exact orthogonal representation, but is not a theorem for every loading.

W3. A restricted computational separation for these newly matched loadings

The W2 rate does not automatically have an efficient estimator. To make this distinction precise, let $w_i = w_1 i^{-\beta} > 0$ with fixed $0 < \beta < 1/K$, and set $M = \tau\sqrt{p}$. Choose constants

$$\frac{1}{K} < a < \min\left\{\frac{1}{2}, \frac{1-2\beta}{K-2}\right\}, \qquad 0 < b < a - \frac{1}{K}.$$

For $s = \lfloor p^a \rfloor$ and $D = \lfloor p^b \rfloor$, every polynomial estimator of total degree at most D in the full observed tensor has worst-case squared risk polynomially larger than the unrestricted $\tau^2 w_1^2$ rate of W2.

Finite statement and proof. For a uniform size-s support S, put $x_S = s^{-1/2}\mathbf{1}_S$ and let D_w be the diagonal map with entries $w_p/w_i \le 1$. Use the rank-one alternatives

$$A_S = \tau\lambda(D_w x_S) \otimes x_S^{\otimes(K-1)}, \qquad 0 < \tau\lambda \le M.$$

They have norm at most M and common weighted target $\Delta = \tau\lambda w_p s^{1-K/2}$. All factors are nonnegative, so for two supports $0 \le \langle A_S, A_T\rangle/\tau^2 \le \lambda^2 |S \cap T|^K / s^K$. The full Gaussian Hermite expansion therefore bounds the squared norm of the centered likelihood ratio through degree D by

$$\chi^2_{\le D} \le \sum_{d=1}^{D} \frac{\lambda^{2d}\mathbb{E}H^{Kd}}{d! s^{Kd}} \le \frac{\mu e^\mu B_D}{1 - B_D} = \eta_D^2, \quad \mu = s^2/p, \quad B_D = \frac{\lambda^2 (KD)^K}{s^K} < 1.$$

For completeness, counting maps by their nonempty fibers gives $H^q = \sum_{j=1}^{q} S(q,j)(H)_j$, where $S(q,j)$ counts partitions of q elements into j nonempty sets. We have $S(q,j) \le j^q/j!$ and $\mathbb{E}(H)_j \le \mu^j$. Hence $\mathbb{E}H^q \le q^q(e^\mu - 1) \le \mu e^\mu q^q$. Substitute $q = Kd$ and sum the geometric upper bound to get the display. For a degree-D estimator with worst-case squared risk R, orthogonality of Hermite polynomials and Cauchy–Schwarz give $|\mathbb{E}_Q T - \mathbb{E}_0 T| \le \sqrt{R}\,\eta_D$. The two target means differ by Δ and their estimation biases are each at most $\sqrt{R}$, so

$$R \ge \Delta^2/(2 + \eta_D)^2.$$

Set $\lambda = \sqrt{p}$. Then $\mu \to 0$ and $B_D = O(p^{1-Ka+Kb}) \to 0$, whereas

$$\frac{\Delta}{\tau w_1} \asymp p^{1/2-\beta-a(K-2)/2} \longrightarrow \infty$$

at a polynomial rate by the choice of a. Such an a exists precisely under the stated sufficient condition $\beta < 1/K$. This proves the restricted separation, using the earlier general low-degree argument with a new loading-compatible prior. It includes diffuse power-law loadings with sharp all-budget rates from W2.

This is not a lower bound for all polynomial-time algorithms or all confidence intervals. An interval honest on zero and these alternatives would give the test $\mathbf{1}\{\Delta \in C(Z)\}$ with total error at most $2\alpha + \mathbb{E}_0|C|/\Delta$. Excluding such efficient tests requires an additional computational hypothesis; the polynomial-estimator result alone does not supply one.

P1. Efficient pointwise optimality at strong saturated Tucker means

This result concerns **Tucker mode caps all equal to** r, $1 \le r \le p$. It does not substitute TT bond caps for mode caps. Let

$$A_0 = \sum_{i \in S} a_i E_i, \qquad |S| = r, \quad a = \min_{i \in S} |a_i| > 0, \quad \|A_0\|_F + \tau \le M \le c_0 a\sqrt{r},$$

where $c_0 > 1$ is fixed. The support and signed amplitudes are not supplied to the algorithm. These means saturate every supplied mode cap. There is a polynomial interval honest on the entire unseparated Tucker class such that, in the strong-signal regime specified below,

$$\mathbb{E}_{A_0}|C| \asymp_{K,\alpha,c_0} \tau\sqrt{r/p},$$

and this matches the infimum expected length at A_0 under that *same outer-class honesty requirement*. Both p and r may grow. This is pointwise optimality in rate, not optimal leading constants.

Algorithm and noise budgets. Make the valid artificial split $Z^{(1)} = Z+\tau G'$, $Z^{(2)} = Z-\tau G'$, with independent errors of scale $v = \sqrt{2}\tau$. Set $\delta = \alpha r/(8p)$ and

$$\eta = 2v\sqrt{2\{(p + p^{K-1})\log 9 + \log(2K/\delta)\}}.$$

Use the preceding cap-valid tangent interval with pilot failure budget δ and total failure budget $\alpha/2$. Its validation quantile is $q = z_{1-(\alpha/2-\delta)/2}$. Intersect it with the direct interval on Z at error $\alpha/2$ and the structural range. All choices are valid on the unseparated class. The total reported length is bounded by $2z_{1-\alpha/4}\tau$, including on pilot failure.

A sufficient condition for the pointwise rate is

$$a \geq 4\eta, \qquad a \geq \eta^2\sqrt{p}/\tau.$$

For fixed order/confidence this is satisfied at sufficiently large constant multiple of $\tau p^{K-1/2}$. This requirement is conservative and much stronger than a claim of optimal computational signal strength.

Upper proof. On the simultaneous pilot event, the top-r projector Q_k satisfies $\|Q_k - P_k\|_{op} \leq d = 4\eta/a$, where P_k projects onto the coordinates in S. The cap-valid certificate has each angle bound at most d by Weyl's inequality and $a \geq 4\eta$. Write T_Q and T_P for their first-order product-space projectors. The representation

$$T_Q = \sum_k \bigotimes_{l\neq k} Q_l - (K-1)\bigotimes_l Q_l$$

with an identity in the omitted mode, and product telescoping, gives $\|T_Q - T_P\|_{op} \leq 2K(K-1)d$. For the diagonal loading, $T_P L = p^{-1/2}\sum_{i\in S} E_i$, so

$$\|T_Q L\|_F \leq \sqrt{r/p} + 2K(K-1)d.$$

The cap-valid bias is bounded by $M\sqrt{\binom{K}{2}}d^2 \leq 16c_0\sqrt{\binom{K}{2}}\sqrt{r}\,\eta^2/a$. The assumed amplitude makes this $O(\tau\sqrt{r/p})$. It also gives $d \leq 4\tau/(\eta\sqrt{p}) = O(\sqrt{r/p})$, because $\eta \gtrsim \tau$ and $r \geq 1$. Thus the selected/intersected interval has length $O(\tau\sqrt{r/p})$ on the pilot event. Off that event its length is $O(\tau)$, with probability at most $\delta = O(r/p) \leq O(\sqrt{r/p})$. The expected-length upper bound follows. The dimension-dependent failure budget is essential to this argument; fixed failure probability would leave an unjustified $O(\tau)$ term.

Lower proof. Put $V = r^{-1/2}\sum_{i\in S} E_i$ and compare A_0 with $A_0 + (\tau/2)V$. Both have mode ranks at most r and norms at most M. Their full Gaussian total variation is less than $1/4$ and their target difference is $(\tau/2)\sqrt{r/p}$. The common-target interval argument at base A_0 proves the lower bound. This lower comparison uses the same outer class on which the algorithm is honest.

Computation. Economy SVDs of the $p \times p^{K-1}$ unfoldings require $O(Kp^{K+1})$ arithmetic and $O(p^K + p^2)$ working storage for fixed K. Tensor projection has polynomial cost of the same order. The new code uses economy SVDs; the previous prototype requested full right singular matrices, which could allocate p^{2K-2} entries. Thus the implementation now scales with the explicit tensor rather than that avoidable allocation. It still reads the full tensor and does not claim sublinear access cost.

What these results do and do not resolve

- Efficient high-dimensional minimax computation is resolved here for $r \gtrsim \sqrt{p}$, where G1 gives the sharp all-budget rate. Original fixed-cap diffuse-loading weak signals remain computationally open.
- All-budget loading rates are resolved for fixed polynomial decay and the specified nonorthogonal perturbations. General profiles remain open; the matching W2 statistical construction still uses ideal block fits.
- General growing-cap adaptation is impossible in the G2 comparison, even between positive cap classes. P1 gives a positive, efficiently attained pointwise rate on a strong saturated Tucker family. It does not characterize pointwise lengths everywhere or optimal SNR thresholds.

No blanket tensor hardness theorem is inferred from low-degree results for sparse tensor PCA. Those results concern a restricted computational model; a testing reduction for arbitrary efficient intervals still needs an explicit, justified computational hypothesis.

Methodological references

- The sparse-functional precursor for G1 is Collier, Comminges and Tsybakov, *Minimax estimation of linear and quadratic functionals on sparsity classes.*
- For the distinct estimation-adaptation problem, see Collier, Comminges, Tsybakov and Verzélen, *Optimal adaptive estimation of linear functionals under sparsity.*
- The restricted computational perspective in W3 follows the methodology of Choo and d'Orsi, *The Complexity of Sparse Tensor PCA.*
- For general tensor-functional tangent inference under signal conditions, see Xu, Chen and Han, *Statistical Inference for Low-Rank Tensor Models.*

The repository-specific dependency and novelty distinctions are recorded in the accompanying audit and literature update.